250721-Classical-PCA-Network-Pipeline

250539-PCA-Next-Step

2025-07-21

Objectives of This Attempt

1. Verify if the mean value of non-zero part of methylation data has effect on partial correlations, including Pearson and Spearman Partial Correlation.
2. Explore the implication of the gap between zeros and minimum non-zero values.

Method:

1. Calculation with Centering with Means of Non-Zero Parts

- Calculate Pearson and Spearman Correlation on Original, Square Rooted, Log Transformed, and INT transformed data
- Compare correlation heatmaps

Correlation changed after the demeaning, don't shfit bulk part

2. Calculation without Centering (Not Centering)

- Subtract the mean of non-zero values from non-zero entries (centering with means) versus no centering
- Repeat calculation and plotting in part 1

For each transformation, take the absolute difference between calculations with centering with means of non-zero parts and not centering. Plot heatmaps of difference graphs.

Loading Data

```
## Load  Methylation Data
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

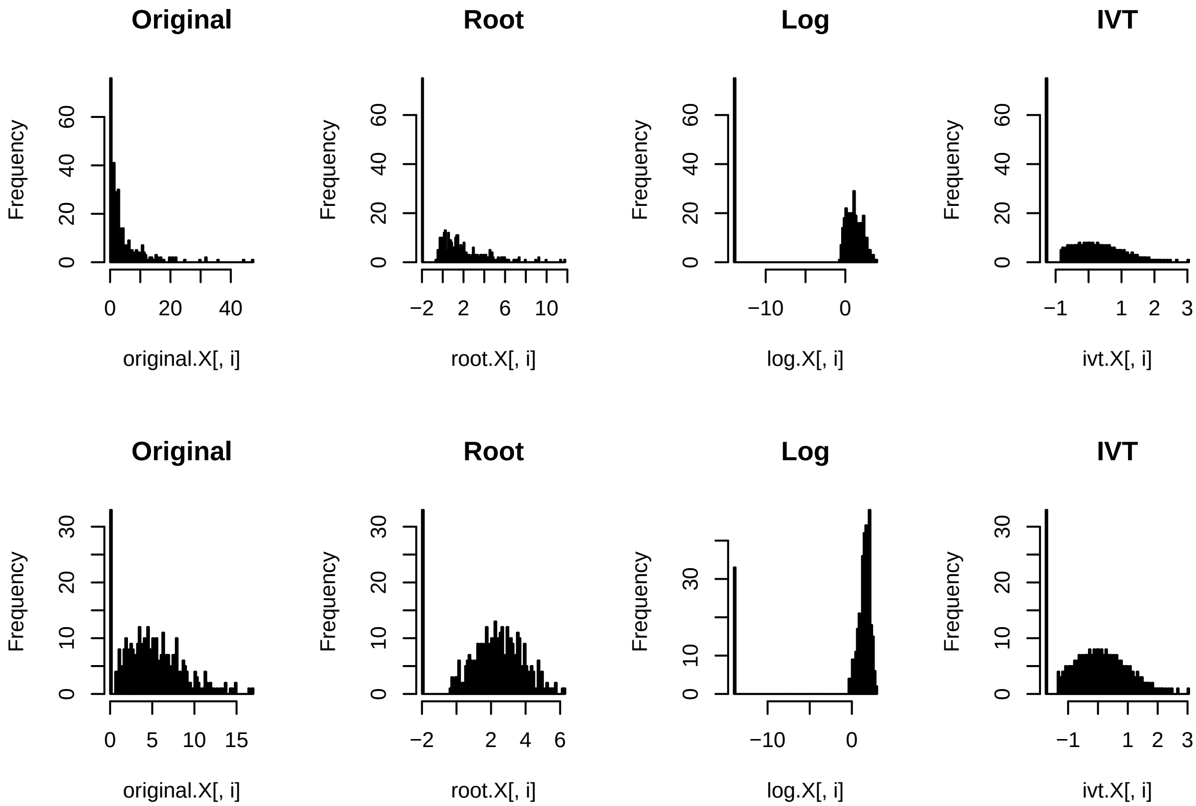
```
## Warning: package 'purrr' was built under R version 4.4.3
```

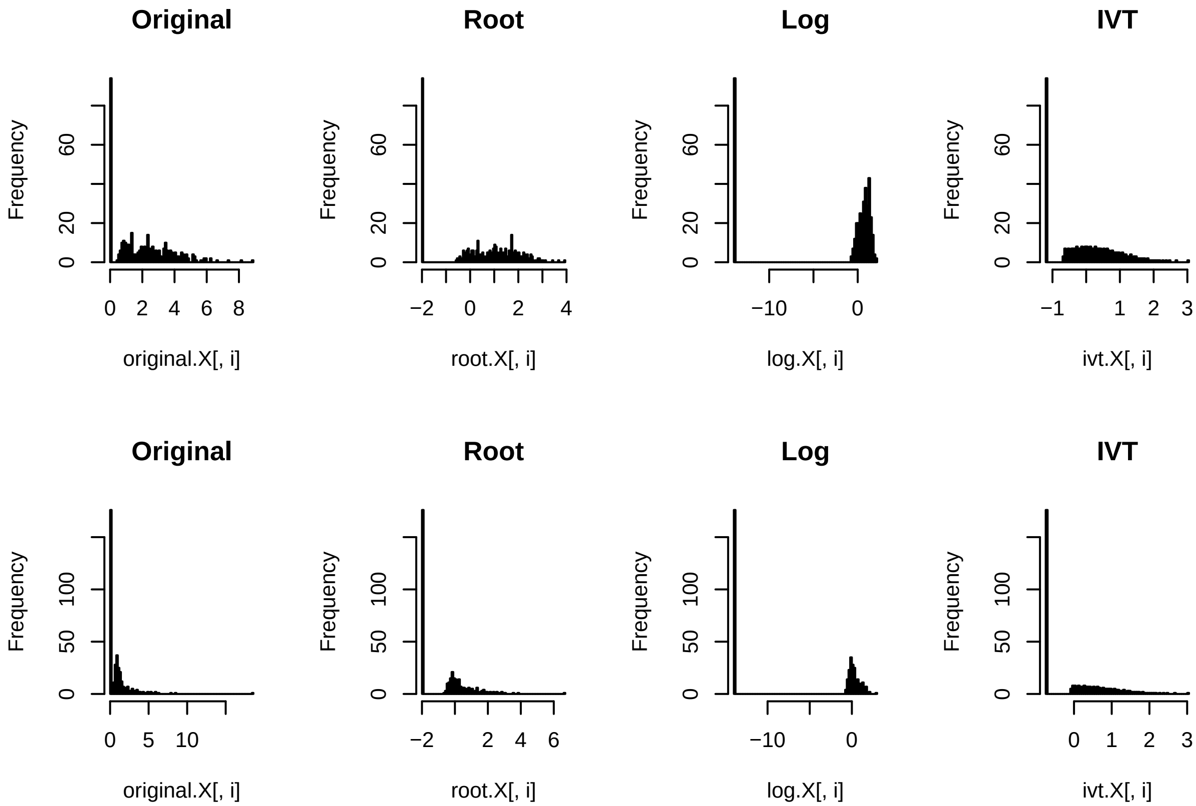
```
X.T <-  readRDS(file = "../../Data/Common_pan_cancer_hyper_bins_adjusted_and_normalized_cnt_in_SE.RDS")
```

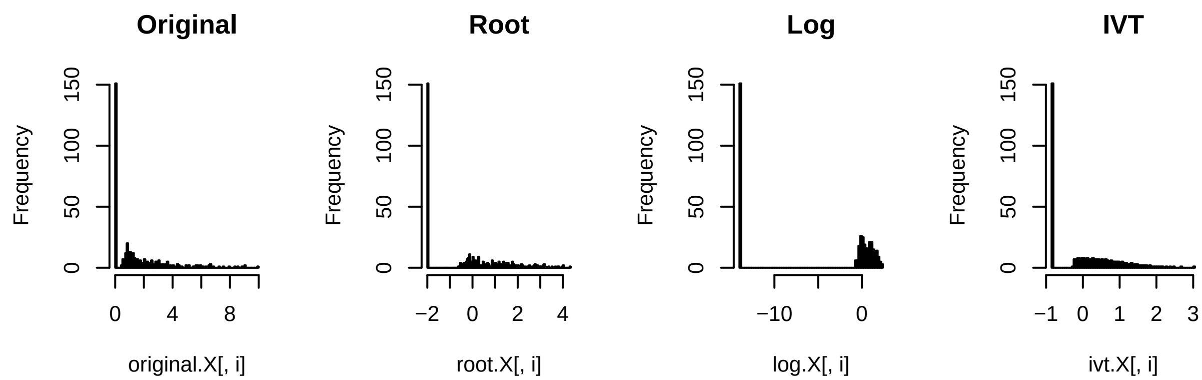
Apply Transformations on Randomly Selected 5 Rows

```r
## Sample 5 DMRs
set.seed(226)
sampDMR <- sample(rownames(X.T), 5)
original.X <- t(X.T[sampDMR, ])

## Box-cox Transformation
source("../R-scripts/IVT.R") ## INT mapps empirical quantiles to normal(0,1) quantiles

## Boxcox applies monotonic transform
BoxCox <- function(X, lambda) {
  apply(X, MARGIN = 2, FUN = function(x) {
    if (lambda == 0) {
      log(x+1e-6)
    } else {
      (x^lambda - 1)/lambda
    }
  })
}

### Root Transform
root.X <- BoxCox(original.X, 0.5)

### Log Transform
log.X <- BoxCox(original.X, 0)


## IVT Transformation
ivt.X <- IVT(original.X, 0.5)
```

```r
## plot distributions of methylations under different transformations,
## One row per DMR
par(mfrow = c(2, 4))
for (i in 1:5) {
  hist(original.X[,i], breaks = 100, main = "Original", xlabs = 'NA', ylabs = 'NA')
  hist(root.X[,i], breaks = 100, main = "Root", xlabs = 'NA', ylabs = 'NA')
  hist(log.X[,i], breaks = 100, main = "Log", xlabs = 'NA', ylabs = 'NA')
  hist(ivt.X[,i], breaks = 100, main = "IVT", xlabs = 'NA', ylabs = 'NA')
}
```


Original
Root
Log
IVT
Frequency
original.X[, i]
root.X[, i]
log.X[, i]
ivt.X[, i]
Original
Root
Log
IVT
Frequency
original.X[, i]
root.X[, i]
log.X[, i]
ivt.X[, i]


Original
Root
Log
IVT
Frequency
0
20
60
0 2 4 6 8
−2 0 2 4
−10 0
−1 1 2 3
original.X[, i]
root.X[, i]
log.X[, i]
ivt.X[, i]
Original
Root
Log
IVT
Frequency
0
50
100
0 5 10
−2 2 4 6
−10 0
0 1 2 3
original.X[, i]
root.X[, i]
log.X[, i]
ivt.X[, i]



Observations

1. The gap between zeros and non-zeros widens as the power transformation `lambda` power approaches to 0.
2. Among power transformations, log effectively transforms non-zero part of variables to be symmetrically distributed near 0.
3. After IVT, the non-zero part of the distribution is close to normally distributed; however, the distribution is truncated normal.

Non-zero mean values effects to Corr

```
library(ppcor)
```

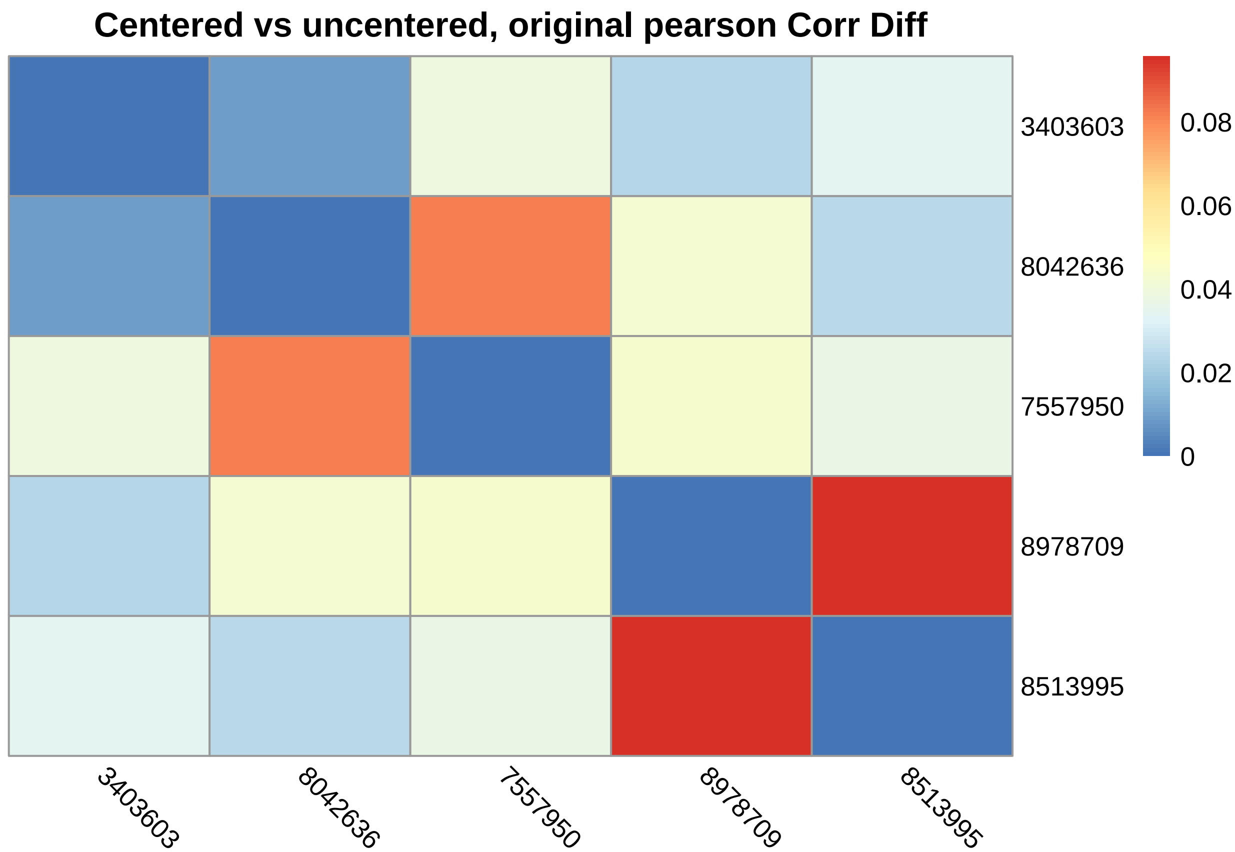
```
## Warning: package 'ppcor' was built under R version 4.4.3
```

```
## Loading required package: MASS
```

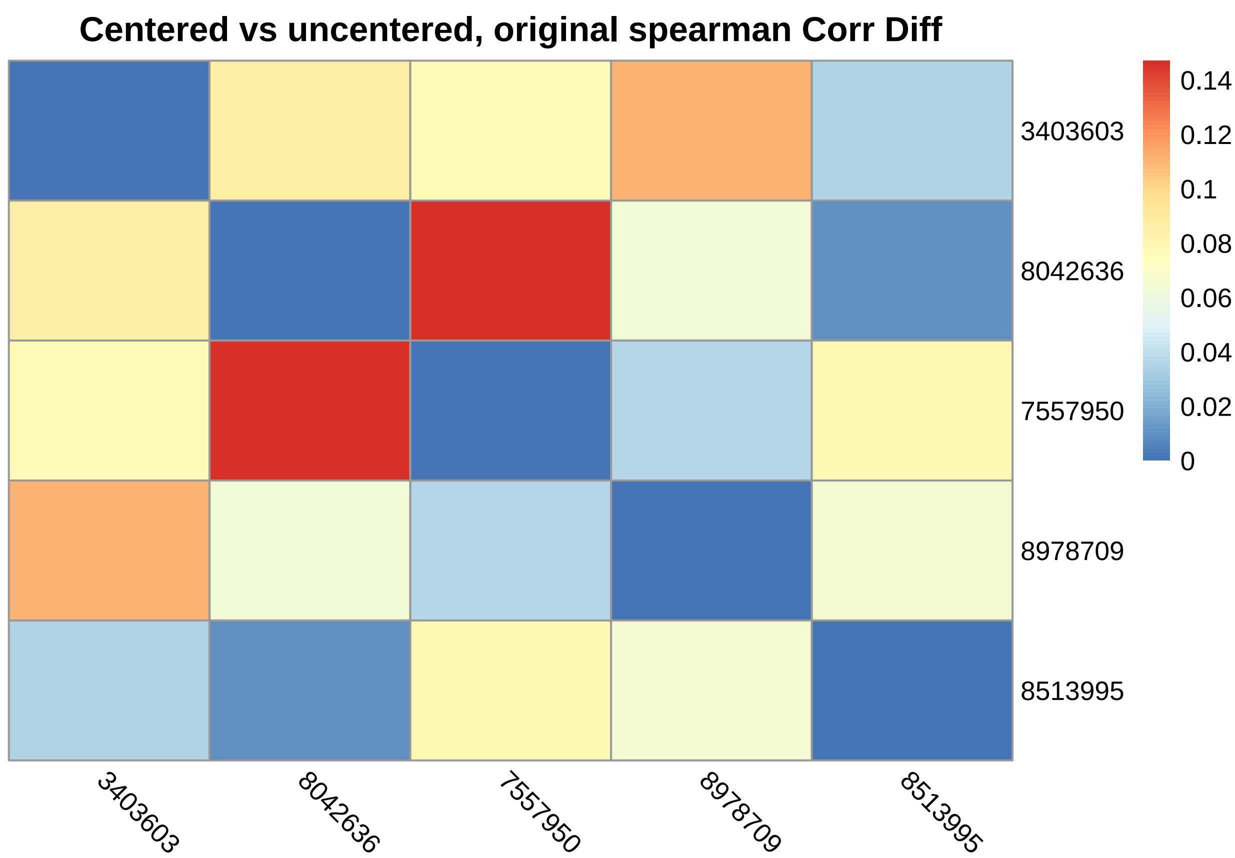
```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

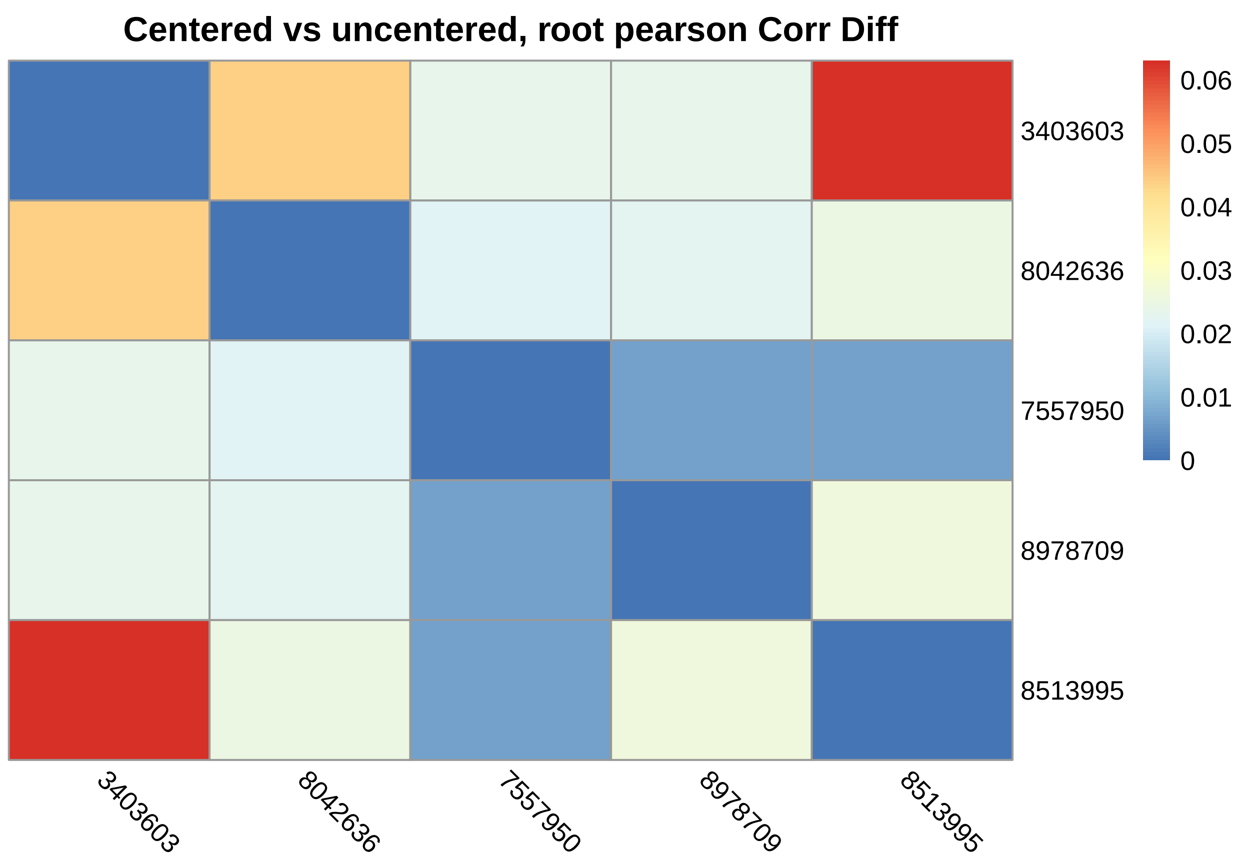
```r
library(pheatmap)

non_min_demean <- function(X) {
  ## For each column, subtract mean value from nonzero values
  ## Keep zero values the same
  apply(X, MARGIN = 2, FUN = function(x) {
    non_zero_indx <- x > min(x)
    non_zero_mean <- mean(x[x > 0])
    x[non_zero_indx] = x[non_zero_indx] - non_zero_mean
    x
  }, simplify = TRUE)
}

original.X.nc <- original.X ## Non-zero part not centered
original.X.c <- non_min_demean(original.X.nc)

pearson_original_nc <- pcor(original.X.nc, method = "pearson")
spearman_original_nc <- pcor(original.X.nc, method = "spearman")

pearson_original_c <- pcor(original.X.c, method = "pearson")
spearman_original_c <- pcor(original.X.c, method = "spearman")

pheatmap(abs(pearson_original_c$estimate -
               pearson_original_nc$estimate),
         cluster_rows = F,  cluster_cols = F,
         angle_col = "315",
         main = "Centered vs uncentered, original pearson Corr Diff")
```



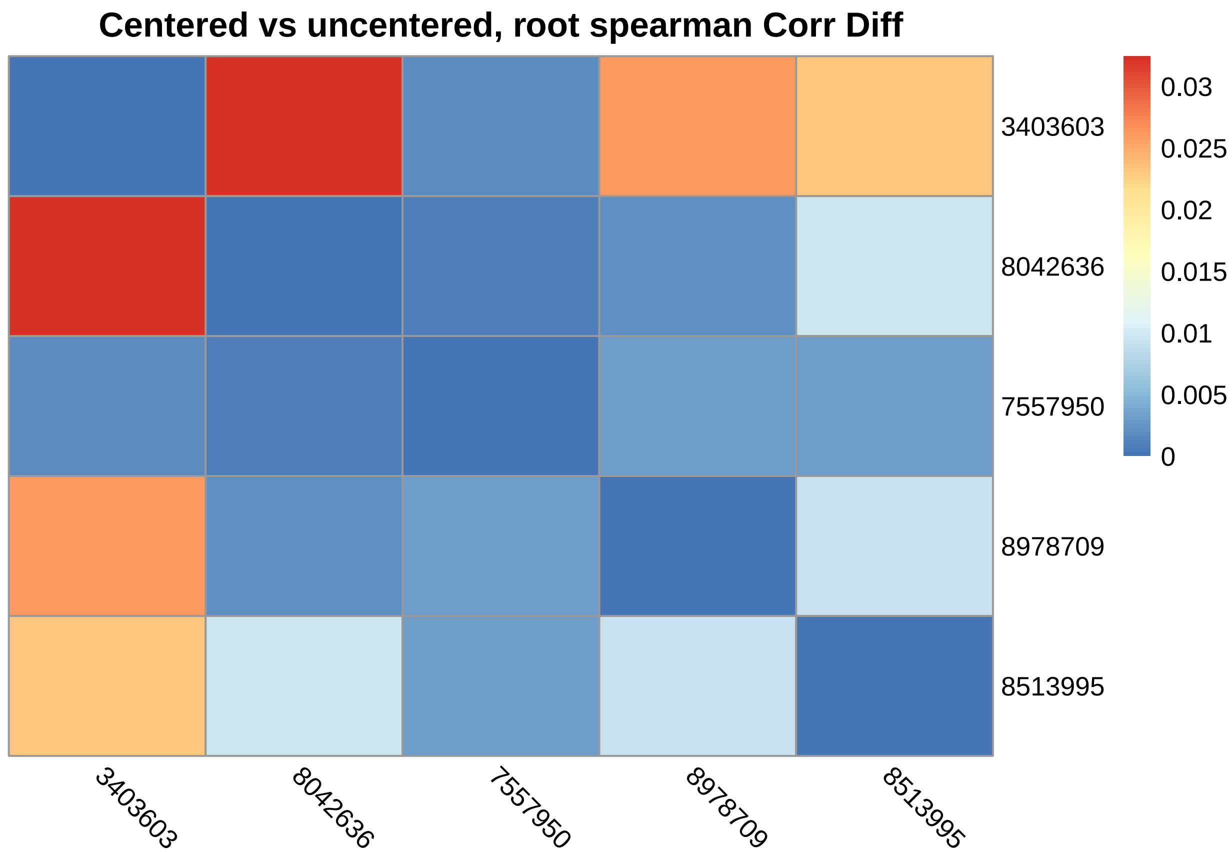

```r
pheatmap(abs(spearman_original_c$estimate -
               spearman_original_nc$estimate),
         cluster_rows = F,  cluster_cols = F,
         angle_col = "315",
         main = "Centered vs uncentered, original spearman Corr Diff")
```



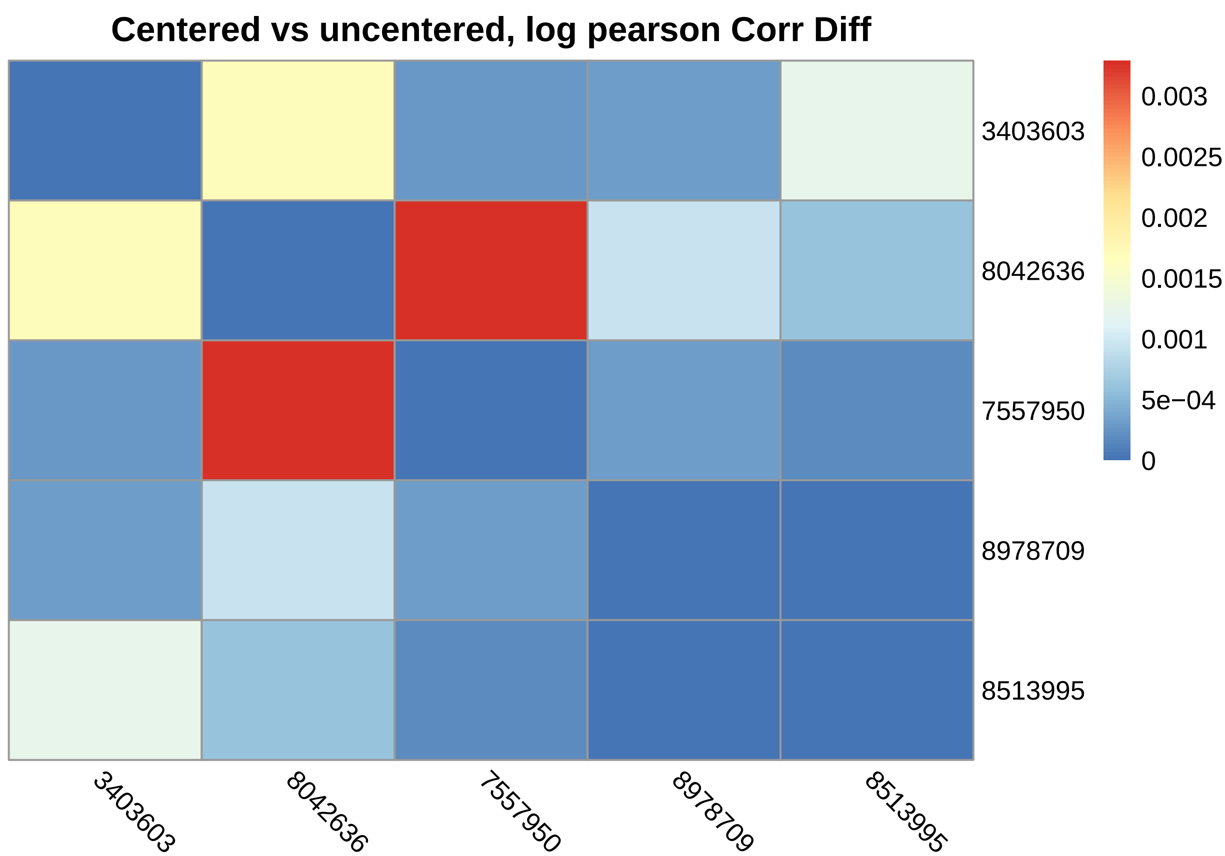

```
root.X.nc <- root.X ## Non-zero part not centered
root.X.c <- non_min_demean(root.X.nc)

pearson_root_nc <- pcor(root.X.nc, method = "pearson")
spearman_root_nc <- pcor(root.X.nc, method = "spearman")

pearson_root_c <- pcor(root.X.c, method = "pearson")
spearman_root_c <- pcor(root.X.c, method = "spearman")

pheatmap(abs(pearson_root_c$estimate -
               pearson_root_nc$estimate),
         cluster_rows = F,  cluster_cols = F,
         angle_col = "315",
         main = "Centered vs uncentered, root pearson Corr Diff")
```




```r
pheatmap(abs(spearman_root_c$estimate -
               spearman_root_nc$estimate),
         cluster_rows = F,  cluster_cols = F,
         angle_col = "315",
         main = "Centered vs uncentered, root spearman Corr Diff")
```



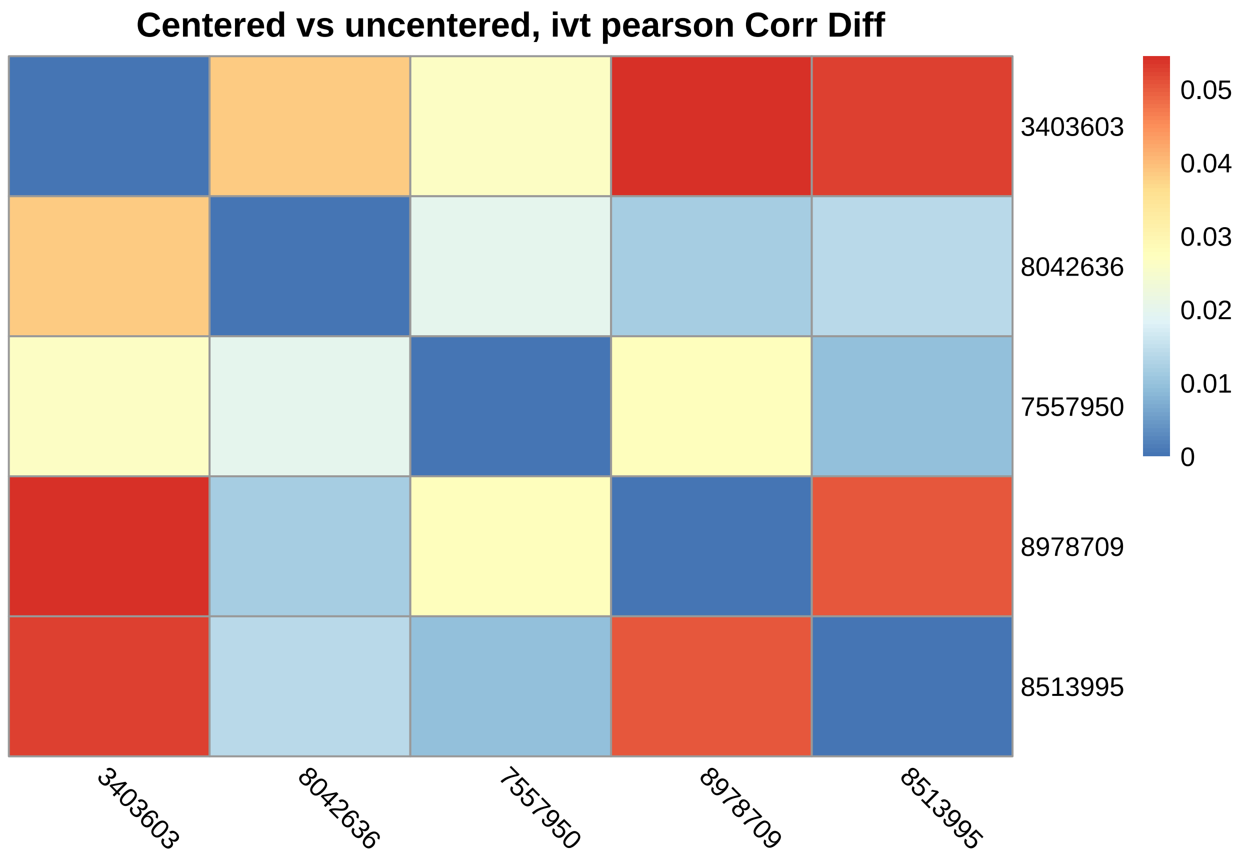

```
log.X.nc <- log.X ## Non-zero part not centered
log.X.c <- non_min_demean(log.X.nc)

pearson_log_nc <- pcor(log.X.nc, method = "pearson")
spearman_log_nc <- pcor(log.X.nc, method = "spearman")

pearson_log_c <- pcor(log.X.c, method = "pearson")
spearman_log_c <- pcor(log.X.c, method = "spearman")

pheatmap(abs(pearson_log_c$estimate -
               pearson_log_nc$estimate),
         cluster_rows = F,  cluster_cols = F,
         angle_col = "315",
         main = "Centered vs uncentered, log pearson Corr Diff")
```



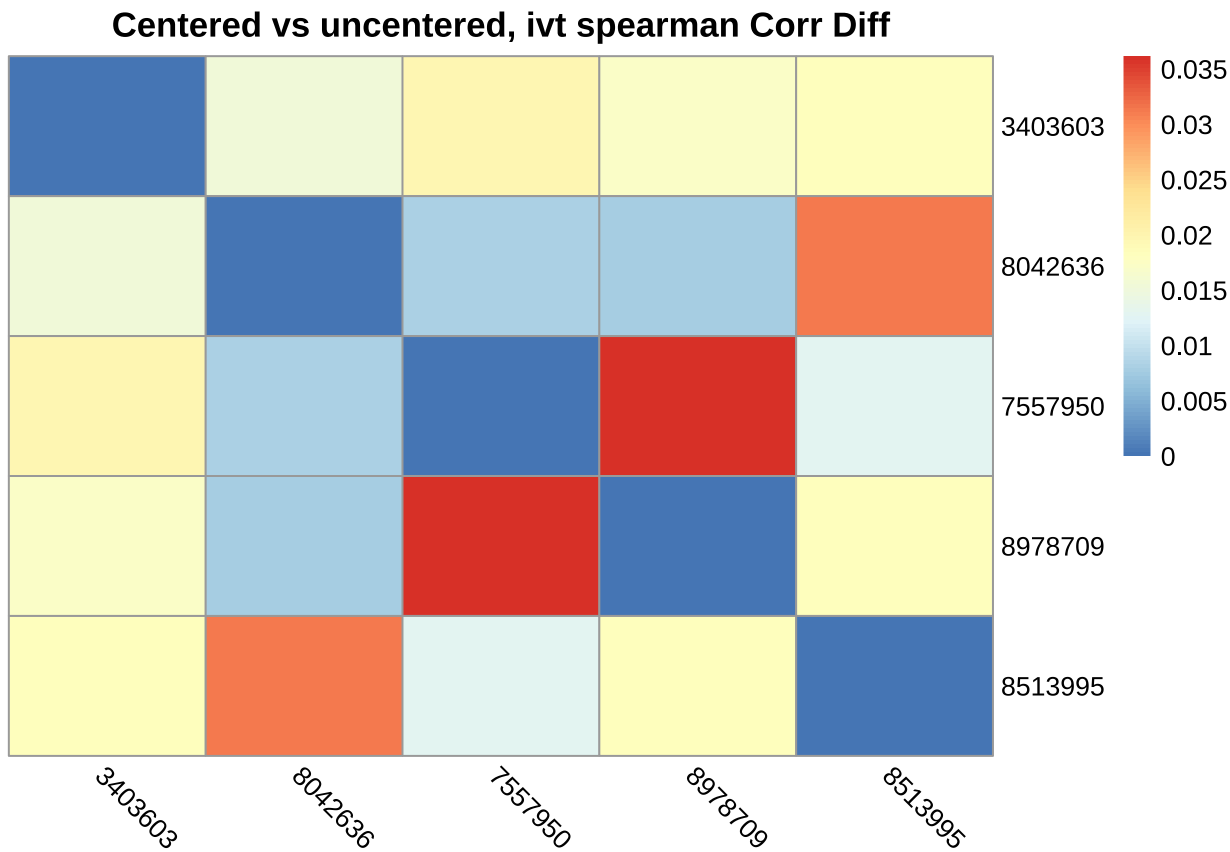

```r
try(
  pheatmap(abs(spearman_log_c$estimate -
                spearman_log_nc$estimate),
         cluster_rows = F,  cluster_cols = F,
         angle_col = "315",
         main = "Centered vs uncentered, log spearman Corr Diff")
)
```

```
## Error in cut.default(x, breaks = breaks, include.lowest = T) :
##   'breaks' are not unique
```

```r
ivt.X.nc <- ivt.X ## Non-zero part not centered
ivt.X.c <- non_min_demean(ivt.X.nc)

pearson_ivt_nc <- pcor(ivt.X.nc, method = "pearson")
spearman_ivt_nc <- pcor(ivt.X.nc, method = "spearman")

pearson_ivt_c <- pcor(ivt.X.c, method = "pearson")
spearman_ivt_c <- pcor(ivt.X.c, method = "spearman")

pheatmap(abs(pearson_ivt_c$estimate -
               pearson_ivt_nc$estimate),
         cluster_rows = F,  cluster_cols = F,
         angle_col = "315",
         main = "Centered vs uncentered, ivt pearson Corr Diff")
```




```r
pheatmap(abs(spearman_ivt_c$estimate -
               spearman_ivt_nc$estimate),
         cluster_rows = F,  cluster_cols = F,
         angle_col = "315",
         main = "Centered vs uncentered, ivt spearman Corr Diff")
```



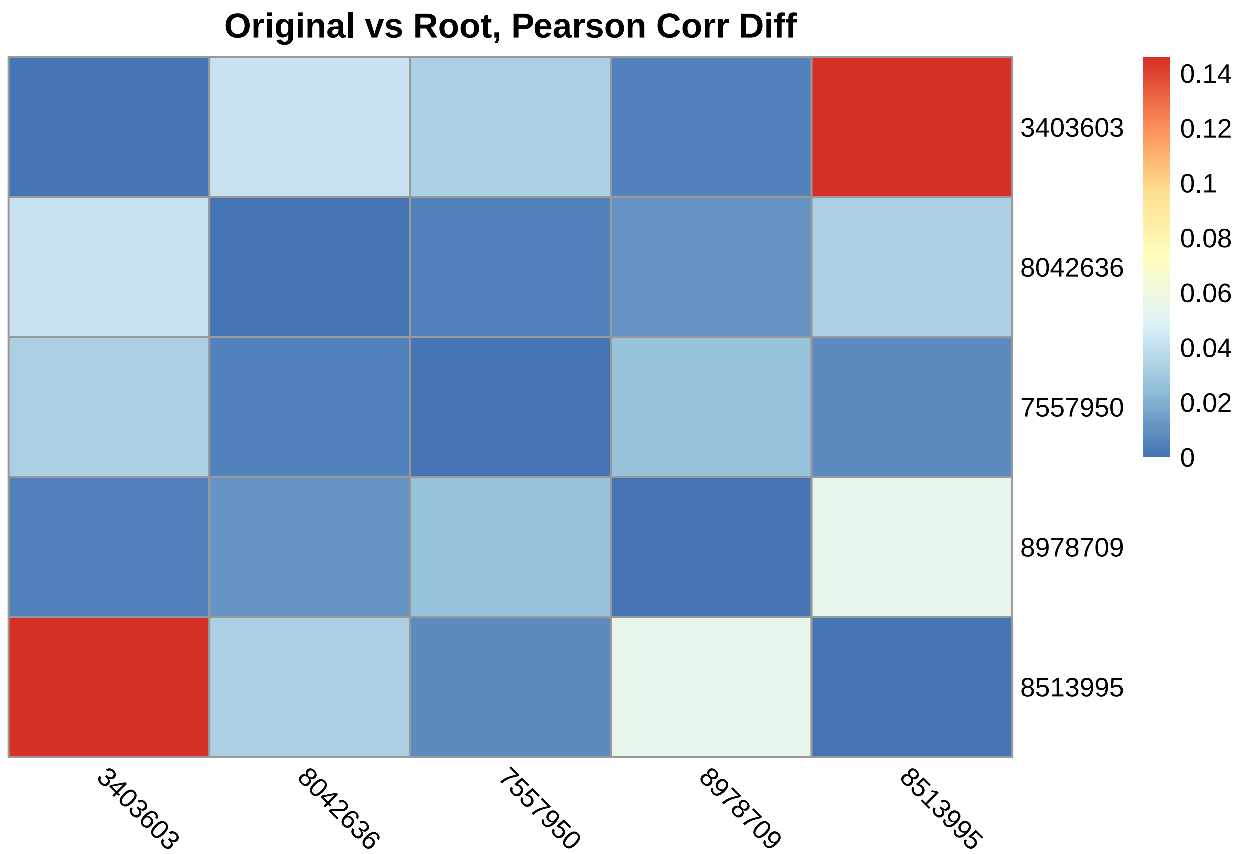
Observation 1. Exclusion of zeros affect Pearson partial correlation more than Spearman partial correlation. 2. Affected DMR pairs were different between Pearson and Spearman

Correlations Change Between Transformations (without Mean Removed)

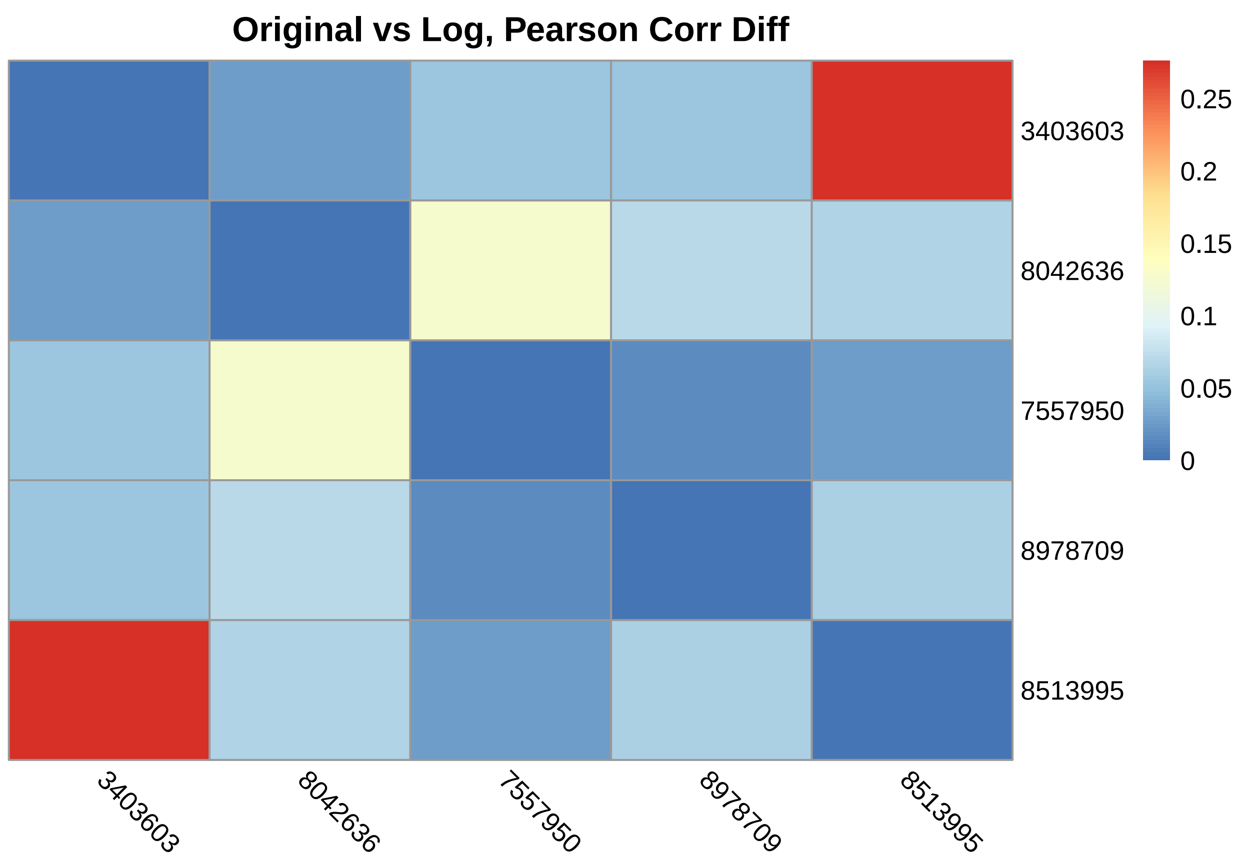
```r
pearson_original_pcor <- pcor(original.X, method = "pearson")
spearman_original_pcor <- pcor(original.X, method = "spearman")

pearson_root_pcor <- pcor(root.X, method = "pearson")
spearman_root_pcor <- pcor(root.X, method = "spearman")

pearson_log_pcor <- pcor(log.X, method = "pearson")
spearman_log_pcor <- pcor(log.X, method = "spearman")

pearson_ivt_pcor <- pcor(ivt.X, method = "pearson")
spearman_ivt_pcor <- pcor(ivt.X, method = "spearman")
```

```r
## Taking differences in partial correlations
pearson_diff_org_root <- abs(pearson_root_pcor$estimate - pearson_original_pcor$estimate)
pearson_diff_org_log <- abs(pearson_log_pcor$estimate - pearson_original_pcor$estimate)
pearson_diff_org_ivt <- abs(pearson_ivt_pcor$estimate - pearson_original_pcor$estimate)

spearman_diff_org_root <- abs(spearman_root_pcor$estimate - spearman_original_pcor$estimate)
spearman_diff_org_log <- abs(spearman_log_pcor$estimate - spearman_original_pcor$estimate)
```

```r
spearman_diff_org_ivt <- abs(spearman_ivt_pcor$estimate - spearman_original_pcor$estimate)

## Plotting Corr Differences before and after transformations
par(mfrow = c(1,3))
pheatmap(pearson_diff_org_root,  cluster_rows = F,  cluster_cols = F,
         angle_col = "315", main = "Original vs Root, Pearson Corr Diff")
```

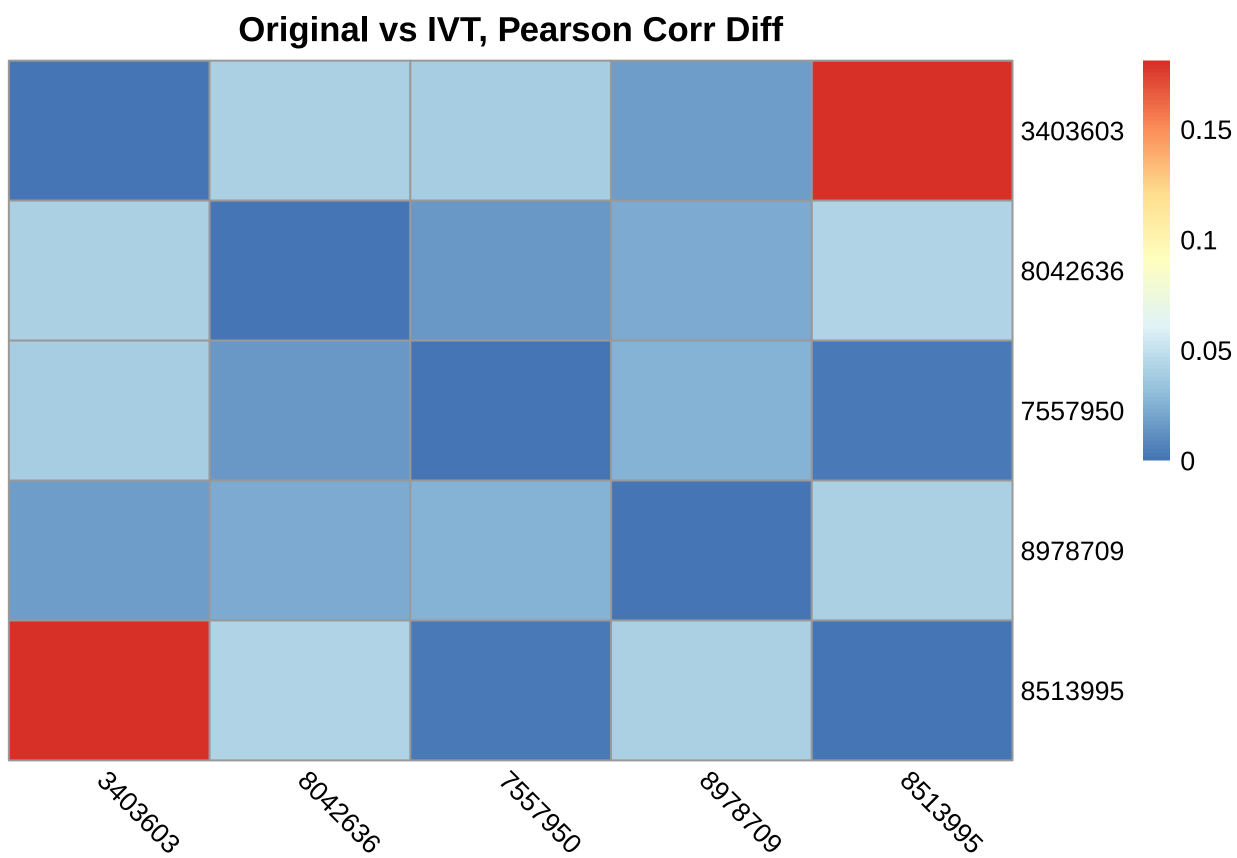



```r
pheatmap(pearson_diff_org_log,  cluster_rows = F,  cluster_cols = F,
         angle_col = "315", main = "Original vs Log, Pearson Corr Diff")
```




```r
pheatmap(pearson_diff_org_ivt,  cluster_rows = F,  cluster_cols = F,
         angle_col = "315", main = "Original vs IVT, Pearson Corr Diff")
```

In graphical lasso, do transformations result in different edges?
Use scatter plot, upper triangular values for partial corr. before and after transformations
How to test significance in the change in partial correlation (optional)




```
try({
  pheatmap(spearman_diff_org_root,  cluster_rows = F,  cluster_cols = F,
         angle_col = "315", main = "Original vs Root, spearman Corr Diff")
  pheatmap(spearman_diff_org_log,  cluster_rows = F,  cluster_cols = F,
           angle_col = "315", main = "Original vs Log, spearman Corr Diff")
  pheatmap(spearman_diff_org_ivt,  cluster_rows = F,  cluster_cols = F,
           angle_col = "315", main = "Original vs IVT, spearman Corr Diff")
})
```

```
## Error in cut.default(x, breaks = breaks, include.lowest = T) :
##   'breaks' are not unique
```

Observations:

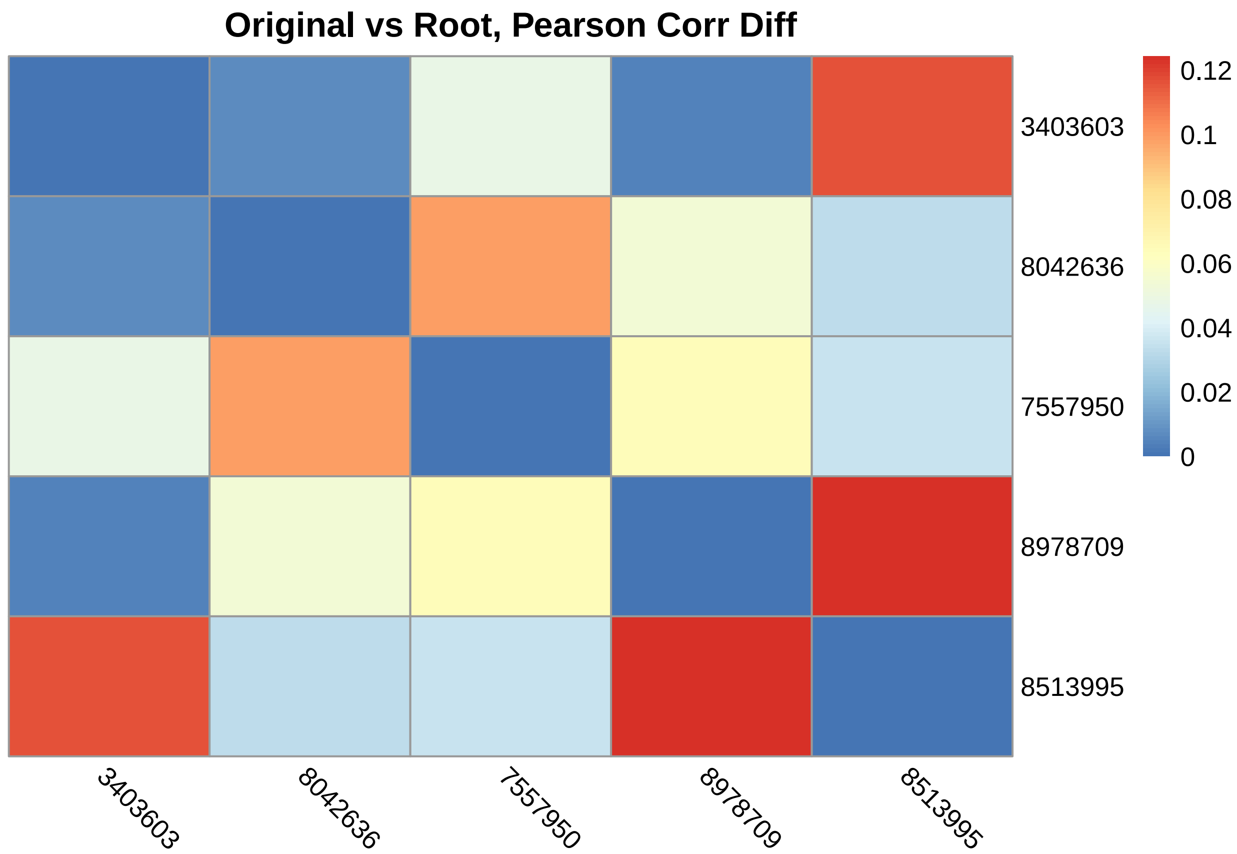
1. Pearson partial correlation witnessed noticeable changes after power and IVT transformations, whereas Spearman didn't due to monotonity in the transformations.
2. In the pearson partial corr, the maximum corr difference range from 0.15 to 0.3, which is big. And such a big change consistently showed up in the same DMR pair.

Correlations Change Between Transformations (with Mean Removed)

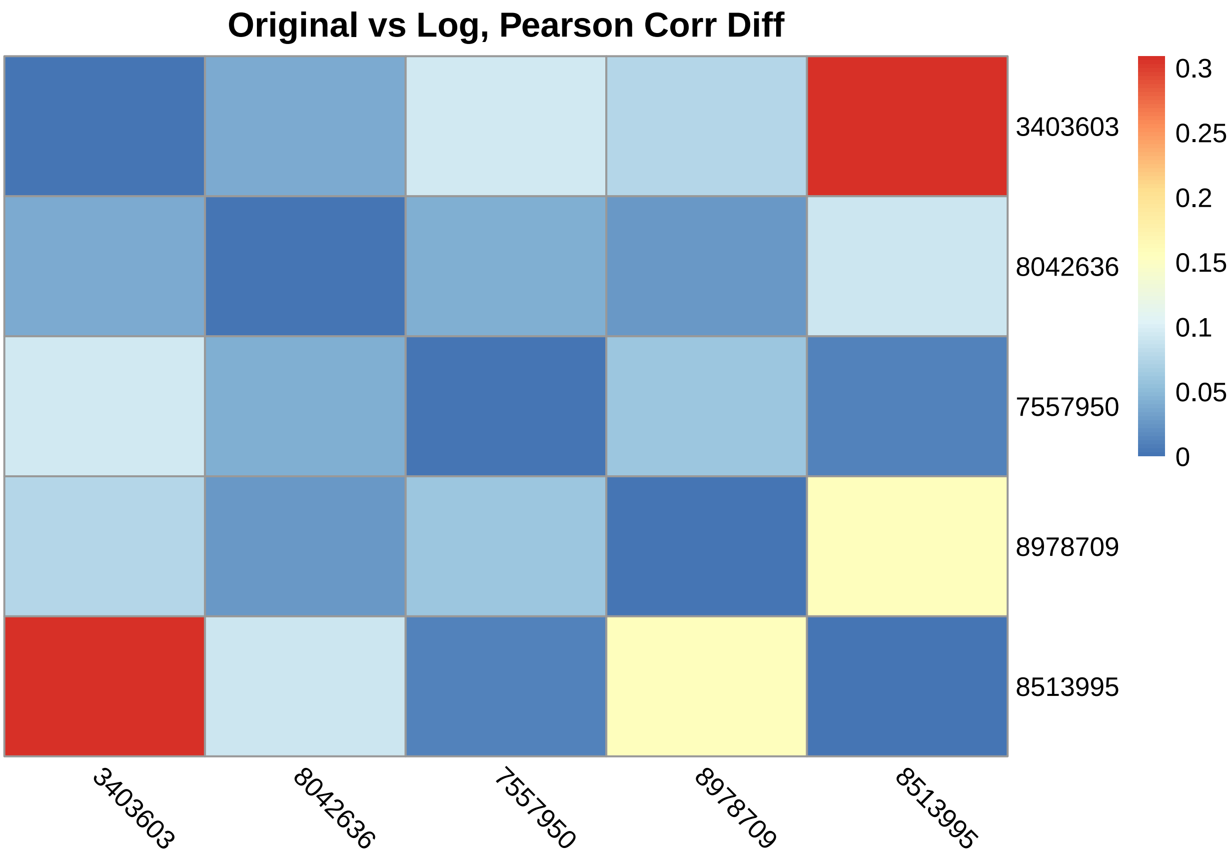
```
## Exclude zeros from calculations
original.X.c <- non_min_demean(original.X)
```

```r
root.X.c <- non_min_demean(root.X)
log.X.c <- non_min_demean(log.X)
ivt.X.c<- non_min_demean(ivt.X)
```

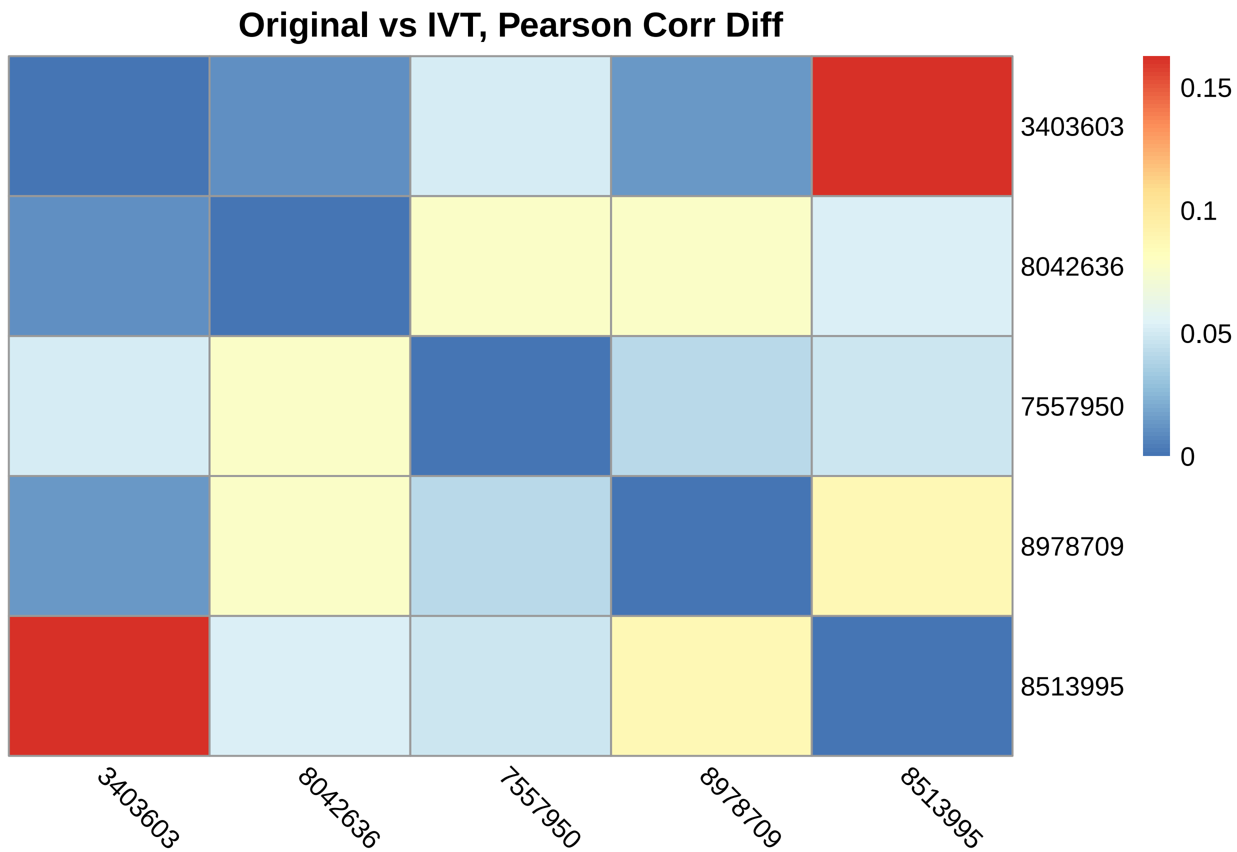
```r
## Repeat the same calculation procedure as before
pearson_original_pcor <- pcor(original.X.c, method = "pearson")
spearman_original_pcor <- pcor(original.X.c, method = "spearman")

pearson_root_pcor <- pcor(root.X.c, method = "pearson")
spearman_root_pcor <- pcor(root.X.c, method = "spearman")

pearson_log_pcor <- pcor(log.X.c, method = "pearson")
spearman_log_pcor <- pcor(log.X.c, method = "spearman")

pearson_ivt_pcor <- pcor(ivt.X.c, method = "pearson")
spearman_ivt_pcor <- pcor(ivt.X.c, method = "spearman")

## Difference Calculation
pearson_diff_org_root <- abs(pearson_root_pcor$estimate - pearson_original_pcor$estimate)
pearson_diff_org_log <- abs(pearson_log_pcor$estimate - pearson_original_pcor$estimate)
pearson_diff_org_ivt <- abs(pearson_ivt_pcor$estimate - pearson_original_pcor$estimate)

spearman_diff_org_root <- abs(spearman_root_pcor$estimate - spearman_original_pcor$estimate)
spearman_diff_org_log <- abs(spearman_log_pcor$estimate - spearman_original_pcor$estimate)
spearman_diff_org_ivt <- abs(spearman_ivt_pcor$estimate - spearman_original_pcor$estimate)

## Plotting Corr Differences before and after transformations
pheatmap(pearson_diff_org_root,  cluster_rows = F,  cluster_cols = F,
         angle_col = "315", main = "Original vs Root, Pearson Corr Diff")
```



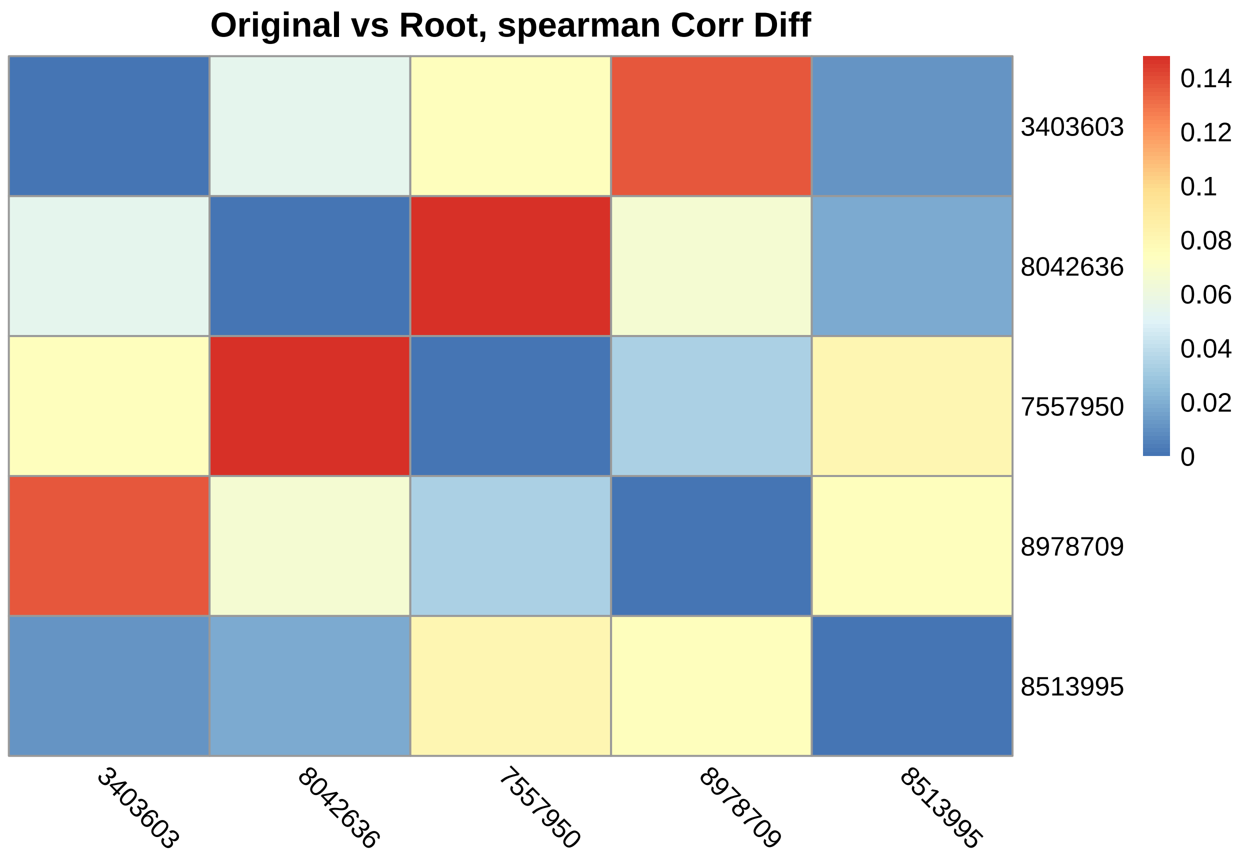

```r
pheatmap(pearson_diff_org_log,  cluster_rows = F,  cluster_cols = F,
         angle_col = "315", main = "Original vs Log, Pearson Corr Diff")
```

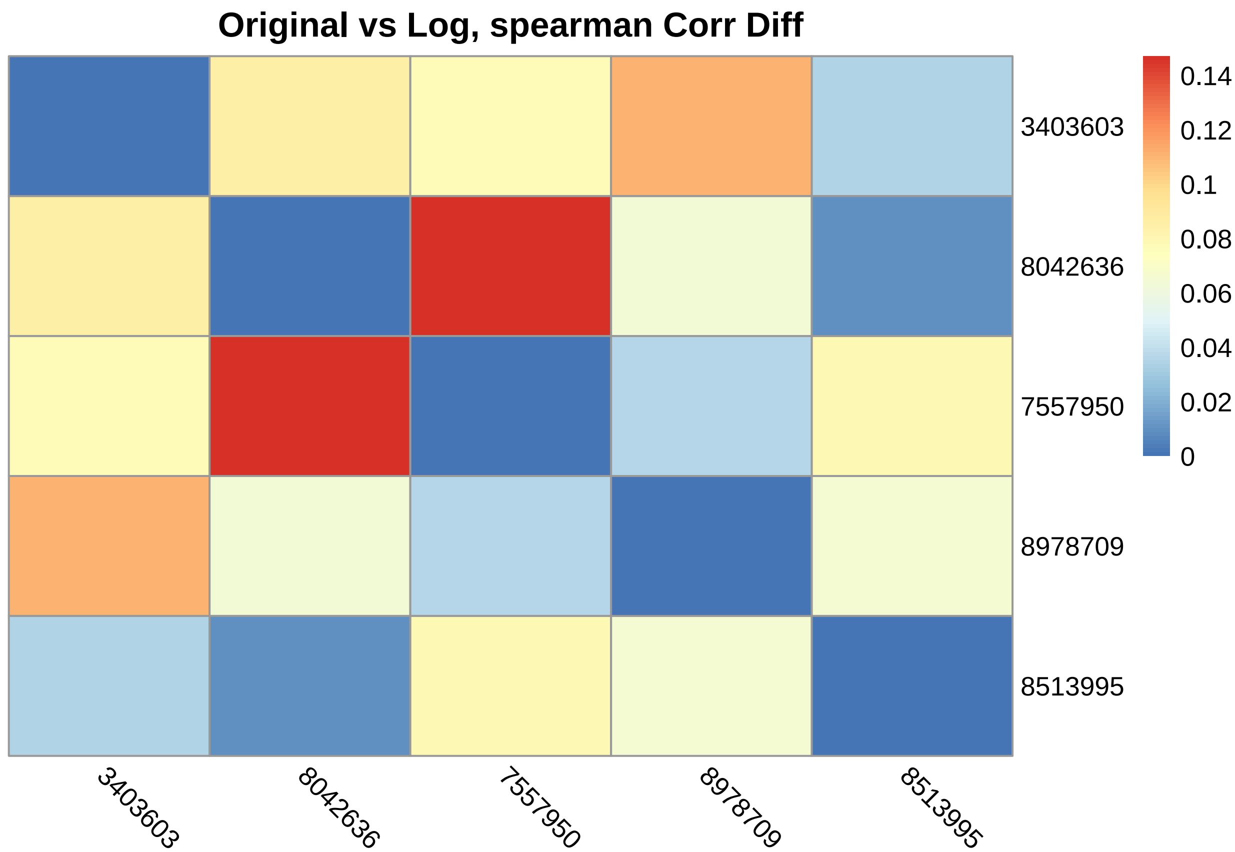


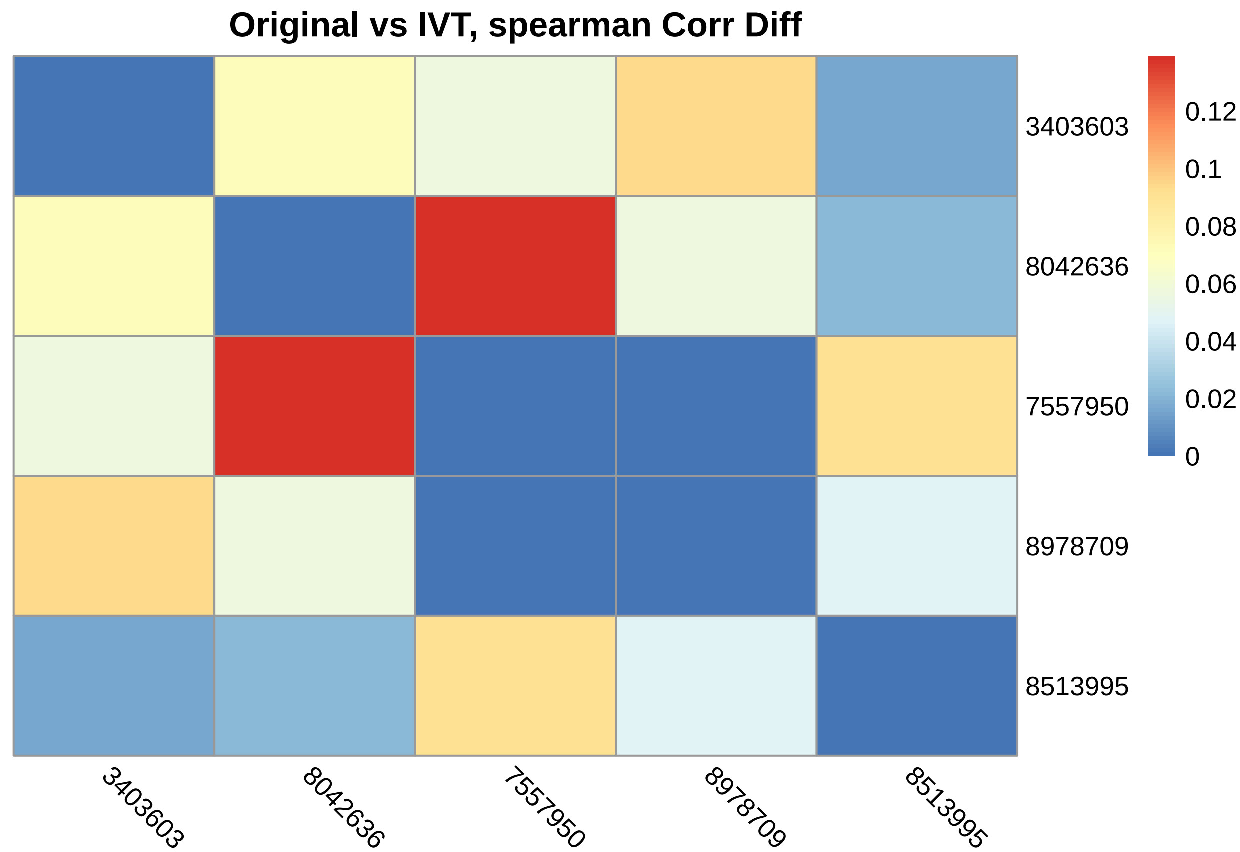

```r
pheatmap(pearson_diff_org_ivt,  cluster_rows = F,  cluster_cols = F,
         angle_col = "315", main = "Original vs IVT, Pearson Corr Diff")
```




```
try({
  pheatmap(spearman_diff_org_root,  cluster_rows = F,  cluster_cols = F,
         angle_col = "315", main = "Original vs Root, spearman Corr Diff")
  pheatmap(spearman_diff_org_log,  cluster_rows = F,  cluster_cols = F,
           angle_col = "315", main = "Original vs Log, spearman Corr Diff")
  pheatmap(spearman_diff_org_ivt,  cluster_rows = F,  cluster_cols = F,
           angle_col = "315", main = "Original vs IVT, spearman Corr Diff")
})
```


Original vs Root, spearman Corr Diff
3403603
8042636
7557950
8978709
8513995
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
3403603
8042636
7557950
8978709
8513995


Original vs Log, spearman Corr Diff
3403603
8042636
7557950
8978709
8513995
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
3403603
8042636
7557950
8978709
8513995



Similar conclusion as without centering the non-zero part

Conclusions

1. Pearson partial correlation is more sensitive to transformation method and exclusion of zeros.
2. With correlation chosen, the change in the correlation before and after transformation is consistent across transformation.
3. Power transformation widens the gap, while IVT doesn't.